



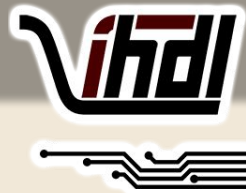
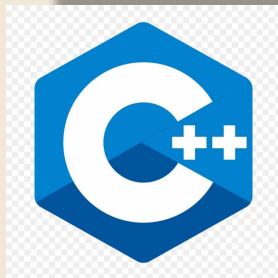
FEE, FIFO, FUM!

WHEN C++ ABSTRACTIONS LIE ABOUT HARDWARE

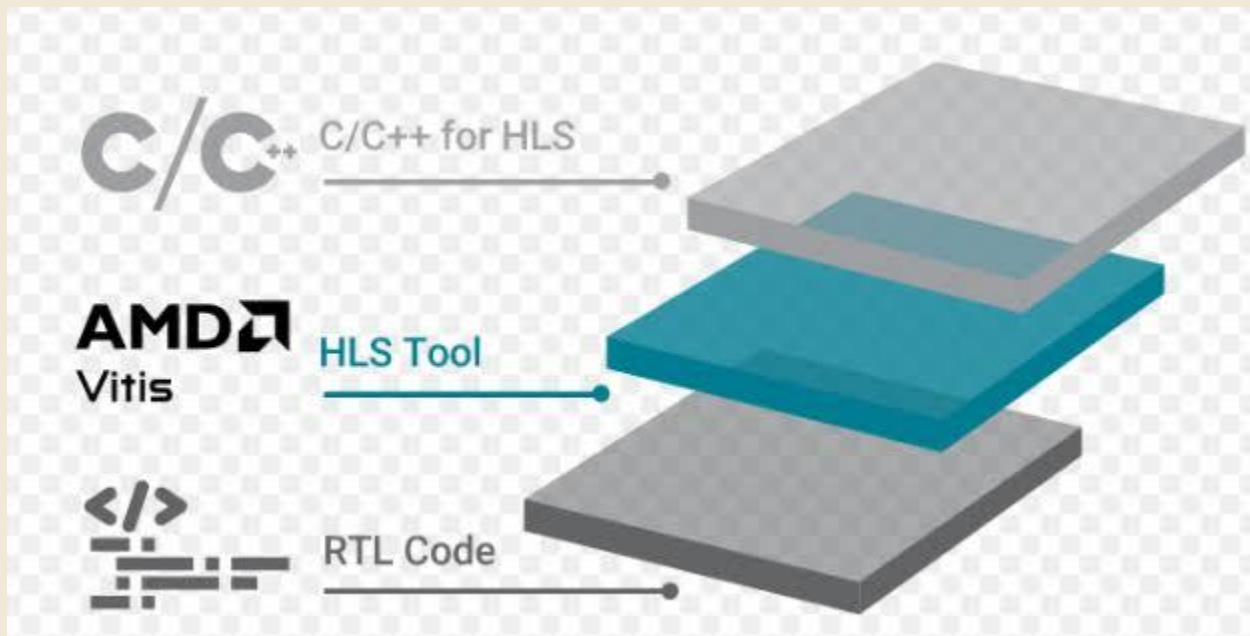
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University of Colorado Boulder
CppCon 2026 Lightning Talks

DESCRIBING HARDWARE



AMD Vitis HLS C++





EXAMPLE SYNCHRONOUS FIFO

SystemVerilog Snippet

```
assign empty = (wr_ptr == rd_ptr);  
  
assign full  = (wr_ptr[PTR_WIDTH]    != rd_ptr[PTR_WIDTH])  
               && (wr_ptr[PTR_WIDTH-1:0] == rd_ptr[PTR_WIDTH-1:0]);
```

HLS Snippets

```
void producer(  
hls::stream<DataWord>& in_raw,  
hls::stream<DataWord>& fifo_stream){  
    if (!in_raw.empty()) {  
        DataWord item = in_raw.read();  
        fifo_stream.write(item);  
    }  
}
```

```
void consumer(  
hls::stream<DataWord>& fifo_stream,  
hls::stream<DataWord>& out_raw){  
    if (!fifo_stream.empty()) {  
        DataWord item = fifo_stream.read();  
        out_raw.write(item);  
    }  
}
```

```
#pragma HLS STREAM variable=internal_fifo depth=16
```



CSimulation: Testbench Logic

```
Compiling ../../../../../../tb_hls_fifo.cpp in debug mode
Compiling ../../../../../../hls_fifo.cpp in debug mode
Generating csim.exe
TEST PASSED (20 words, depth=16)
INFO [HLS SIM]: The maximum depth reached by any hls::stream() instance in the design is 20
INFO: [SIM 211-1] CSim done with 0 errors.
INFO: [SIM 211-3] ***** CSIM finish *****
```

Wait a minute...

CSynthesis: C++ HLS to RTL

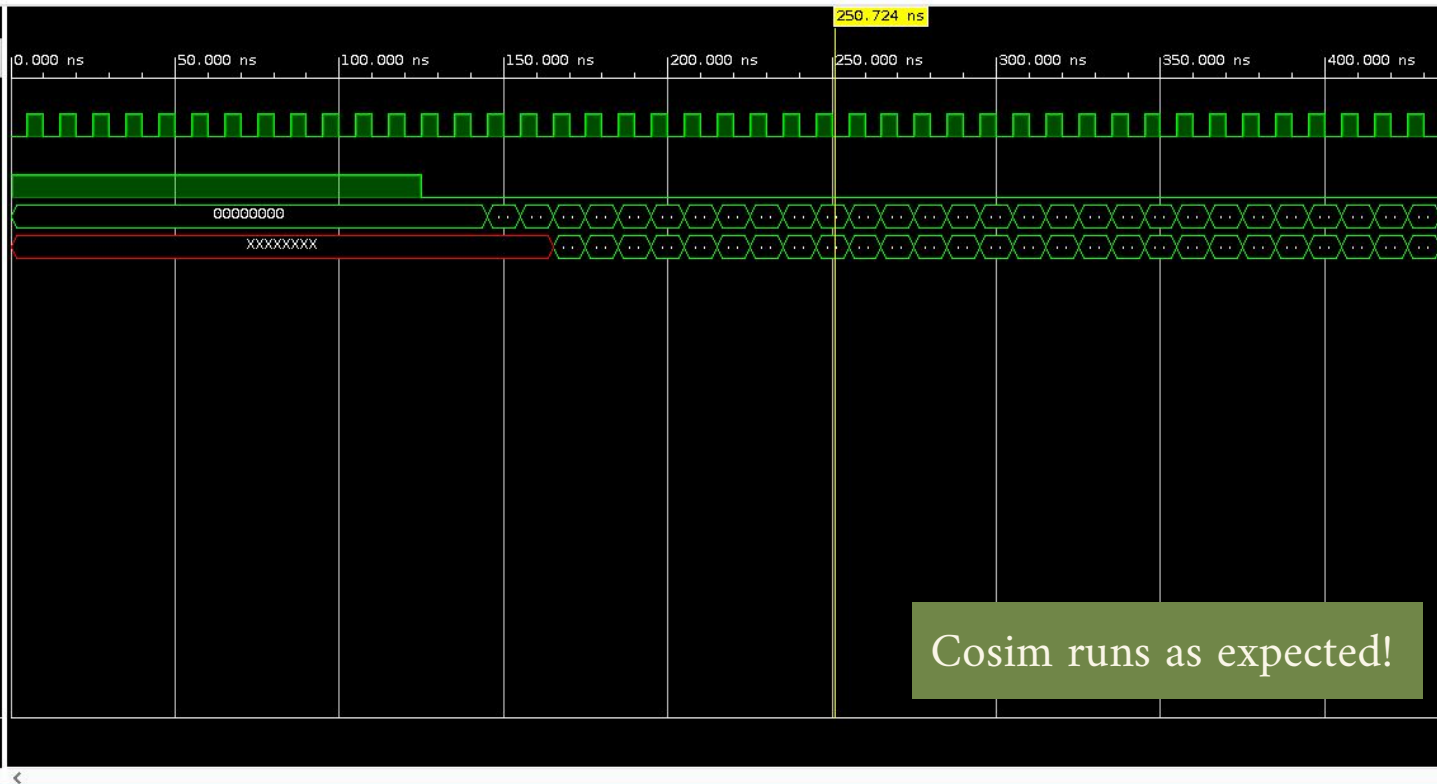
```
[RTGEN 206-100] Finished creating RTL model for 'hls_fifo_top'.  
[RTMG 210-285] Implementing FIFO 'internal_fifo_U(hls_fifo_top_fifo_w32_d16_S)'  
[RTMG 210-285] Implementing FIFO 'start_for_consumer_U0_U(hls_fifo_top_start_for
```

Accurate depth... looking good to me...

hls_fifo_top.wcfg x AESL_deadlock_detector.v x xlnx_uvm_package.sv x



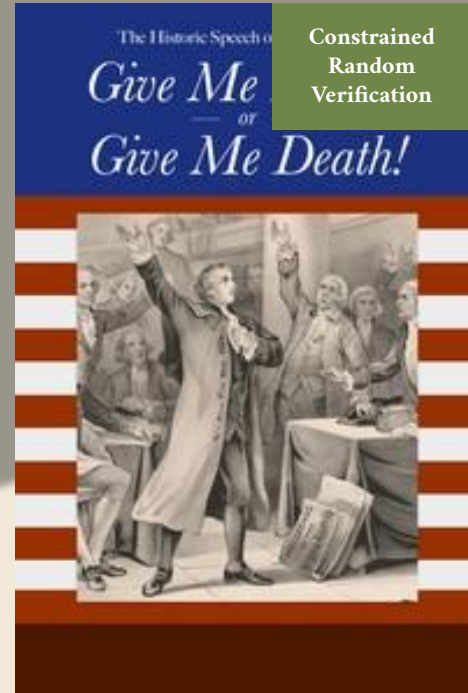
Name	Value
▼ Clock	
ap_clk	0
▼ Reset	
ap_rst	0
> in_r_dout[31:0]	0000000b
> out_r_din[31:0]	00000008

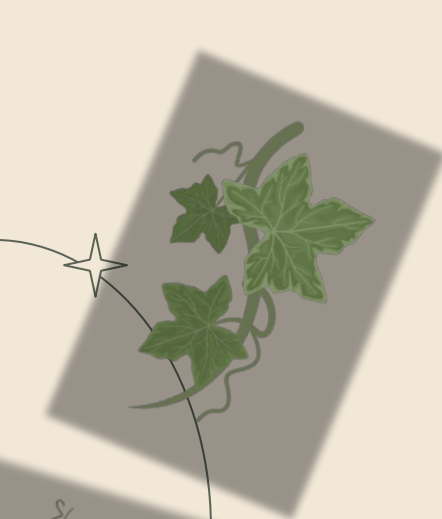


Cosim runs as expected!

MY TAKEAWAY

HLS is fascinating
and useful, but
HDLs are here to
stay!





THANKS!

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